

HOW TO PICK YOUR TEST

What's your data type? → Interval/ratio? → Check normality → ✓ Normal = Parametric

Ordinal? → Non-parametric (ordinal)

Categorical? → Non-parametric (categorical)

Design? → Between groups = different people vs Within groups = same people

● Parametric ● Non-param (ordinal) ● Non-param (categorical) ● Between groups ● Within groups

👉 Click any card to expand

	Between groups Independent samples — different people	Within groups Repeated measures — same people
Parametric (interval, normal) 2 groups	Independent samples t-test Parametric ▲ What it tests Are the means of two independent groups significantly different? Assumptions: Normal distribution in both groups, interval data, homogeneity of variances (Levene's test) Example: Do men and women differ in exam scores? Group A (men) vs Group B (women) — each person tested once. → $t(df) = \dots, p < .05$ Also called: Unrelated t-test, between-subjects t-test	Related (paired) t-test Parametric ▲ What it tests Is there a significant difference in means across two time points or conditions for the same participants? Assumptions: Differences between pairs are normally distributed, interval data Example: Same students tested before and after revision. Score at Time 1 vs Time 2. → $t(df) = \dots, p < .05$ Key idea: Subtracts each person's scores — removes individual variation. More powerful than independent t-test.
Parametric (interval, normal) >2 groups	One-way ANOVA Parametric ▲ What it tests Are there differences in means across 3+ independent groups? Assumptions: Normality (Shapiro-Wilk), homogeneity of variances (Levene's). If variances unequal → <i>Welch ANOVA</i> Example: 3 groups each get a different drug. Compare mean recovery times. $F(2, 87) = 5.3, p = .007, \eta^2 = .11$ Post-hoc needed if F is significant — to find WHICH groups differ (Tukey, Bonferroni).	Repeated measures ANOVA Parametric ▲ What it tests Do means differ across 3+ conditions/timepoints measured on the same people? Assumptions: Normality + <i>Sphericity</i> (Mauchly's test). If violated → Greenhouse-Geisser or Huynh-Feldt correction Example: Same students tested after re-reading, answering questions, and creating questions. $F(2, 10) = 19.09, p < .001, \eta^2 = .79$

Test statistic: $F = MS_{\text{between}} / MS_{\text{within}}$

Sphericity: Assumes variances of differences between all condition pairs are equal (the within-subjects version of homogeneity).

Non-parametric
(ordinal data)
2 groups

Mann-Whitney U

Non-param (ordinal) ▲

What it tests

Do two independent groups have different rank distributions? (Non-parametric alternative to independent t-test)

When to use: Data is ordinal, or normality is violated with small samples

Example: Compare pain rankings (1–10 scale) between two treatment groups. Ranks all scores together, compares rank sums.

Also called: Wilcoxon Rank-Sum test. Tests medians, not means.

Wilcoxon Signed-Rank

Non-param (ordinal) ▲

What it tests

Are there differences between two related conditions in the same people? (Non-parametric alternative to paired t-test)

When to use: Ordinal data, or normality violated in paired design

Example: Participants rate anxiety before and after therapy (ordinal 1–10). Ranks the magnitude of differences, ignores ties.

Difference from sign test: Considers the size of differences, not just direction.

Non-parametric
(ordinal data)
>2 groups

Kruskal-Wallis

Non-param (ordinal) ▲

What it tests

Do 3+ independent groups differ in rank distributions? (Non-parametric alternative to one-way ANOVA)

When to use: Normality violated OR ordinal DV, 3+ groups, different participants

Example: Compare satisfaction ratings across 3 different apps. Ranks all scores, tests if rank sums differ between groups.

Note: Like Mann-Whitney, but for 3+ groups. Still needs post-hoc if significant.

Friedman's ANOVA

Non-param (ordinal) ▲

What it tests

Do the same participants differ across 3+ conditions? (Non-parametric alternative to repeated measures ANOVA)

When to use: Normality violated in repeated measures design, or ordinal data across 3+ timepoints

Example: Same students scored after re-reading, answering questions, creating questions. Ranks each person's scores within their own row, then compares rank sums across conditions.

Test stat: $M = 12 / (Nk(k+1)) \times \sum R_i^2 - 3N(k+1)$

Non-parametric
(categorical data)

Chi-square (χ^2) test

Non-param (categorical) ▲

What it tests

Is there an association between two categorical variables in independent

Binomial sign test

Non-param (categorical) ▲

What it tests

In a repeated/matched design, do participants systematically score higher

groups? Are observed frequencies different from expected?

Assumptions: Expected frequency ≥ 5 in each cell, independent observations

Example: Is there a relationship between gender (M/F) and voting choice (A/B/C)?

$\rightarrow \chi^2(2) = 8.3, p = .016, \text{Cramér's } V = .21$

Chi-square = $\sum (O - E)^2 / E$ — bigger = more different from expected.

in one condition vs another? (Weakest non-param test — only uses direction, not magnitude)

When to use: Categorical outcome or when you can only tell if scores went up or down, not by how much

Example: 20 students tested before/after tutoring. 15 improved, 5 worsened. Is 15/20 significantly above chance (50%)? Uses binomial distribution.

Weaker than Wilcoxon: Wilcoxon uses the size of differences; sign test only uses +/-. Use sign test when you can't rank differences.

QUICK REFERENCE — PARAMETRIC ↔ NON-PARAMETRIC PAIRS

Parametric	Non-param ordinal alternative	Design
Independent t-test	Mann-Whitney U	Between / 2 groups
Paired t-test	Wilcoxon Signed-Rank	Within / 2 groups
One-way ANOVA	Kruskal-Wallis	Between / 3+ groups
Repeated measures ANOVA	Friedman's ANOVA	Within / 3+ groups
(no parametric equiv.)	Chi-square / Sign test	Categorical data